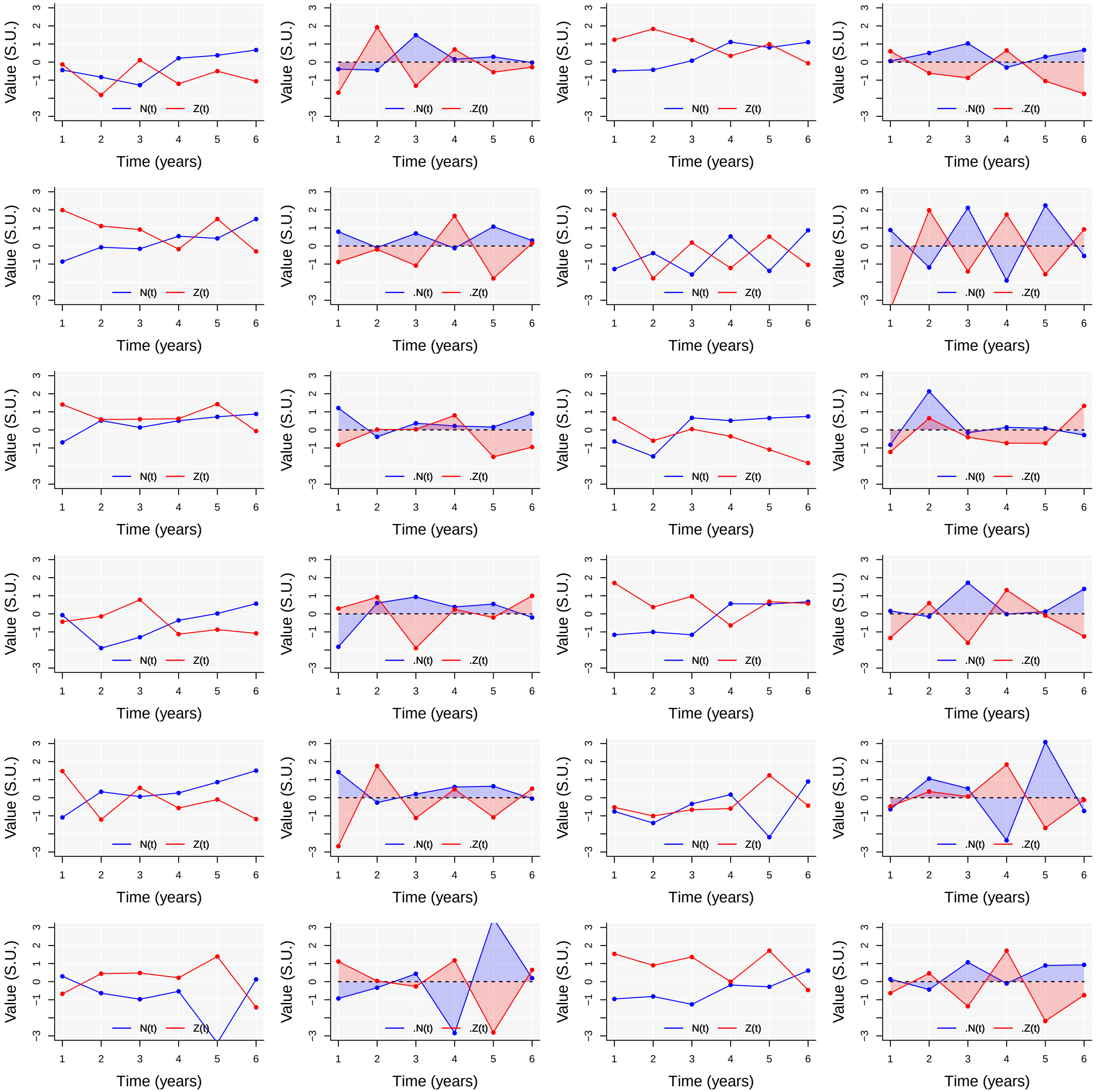
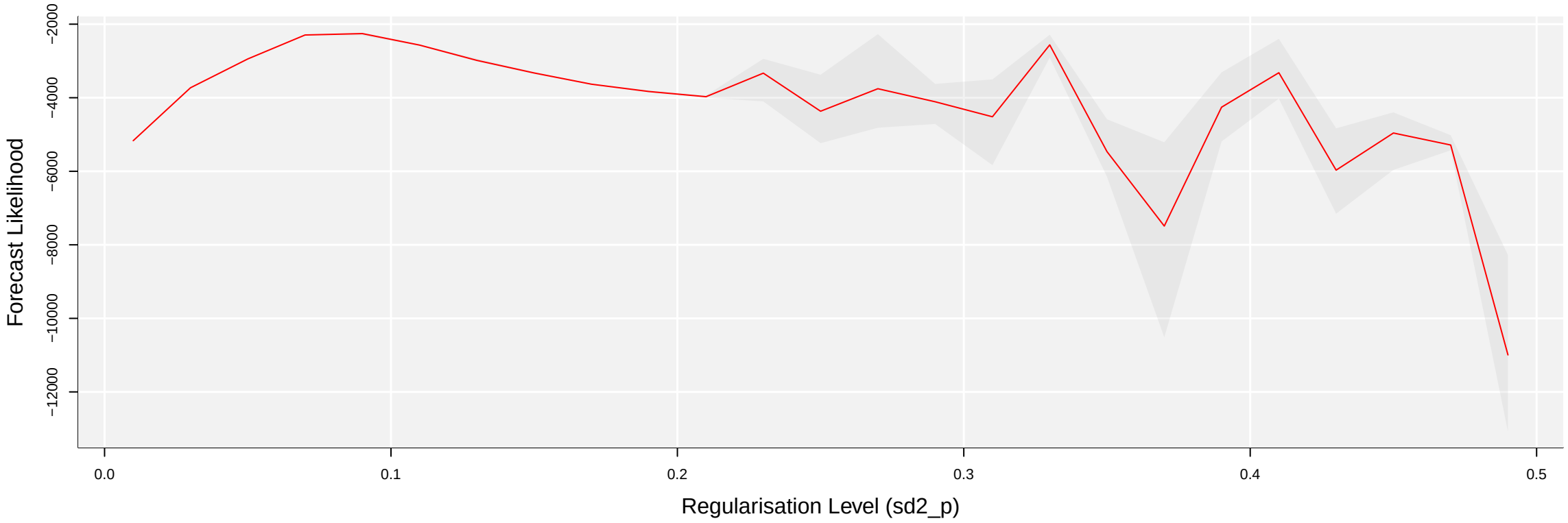
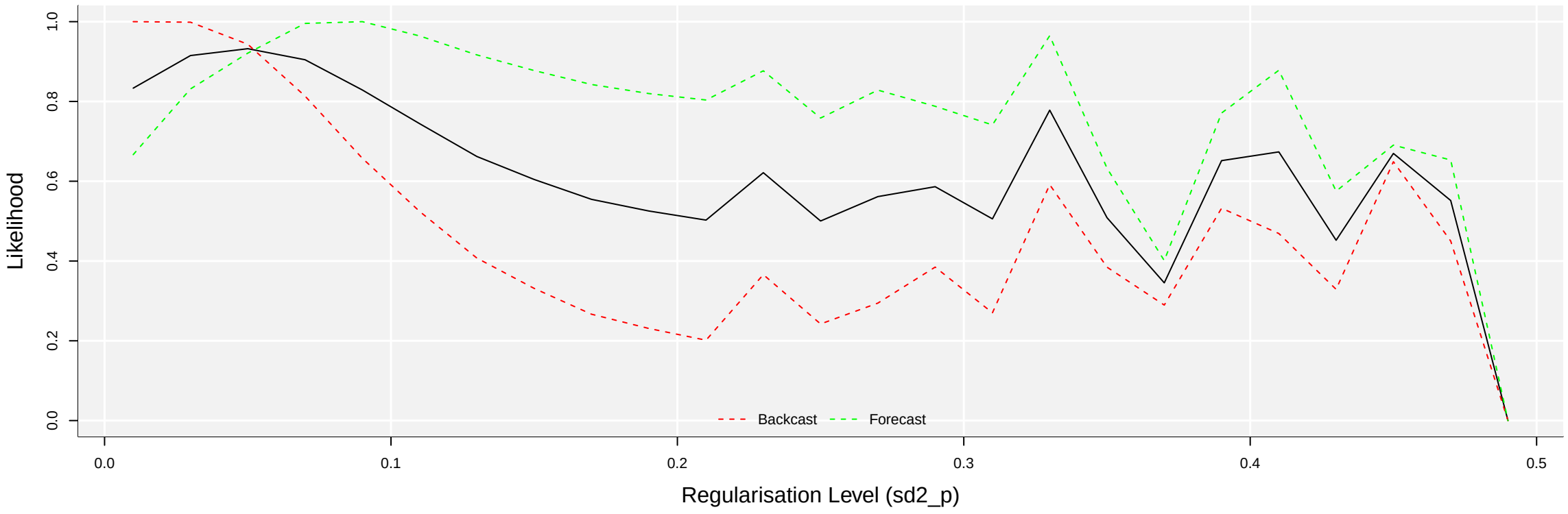
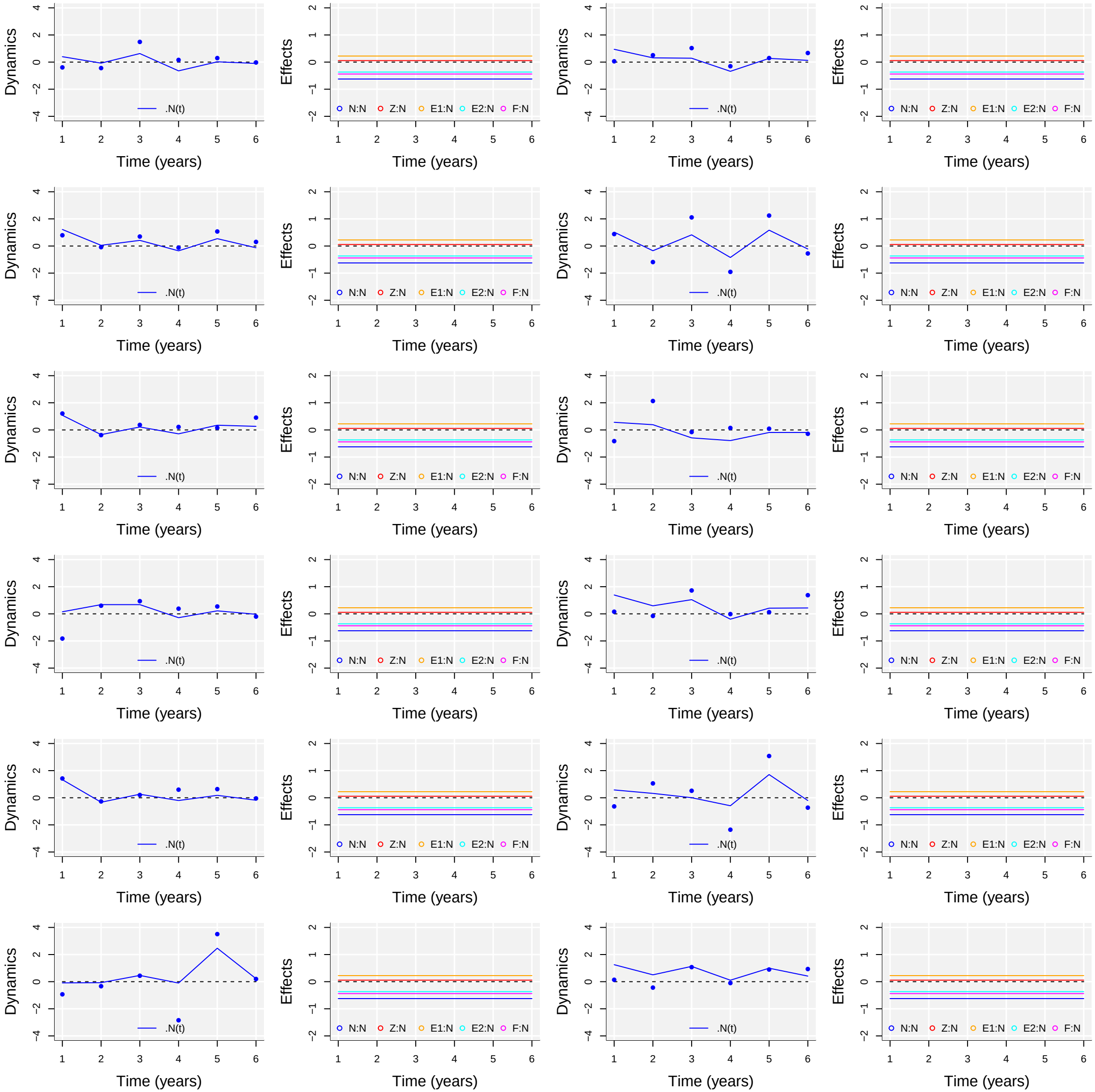
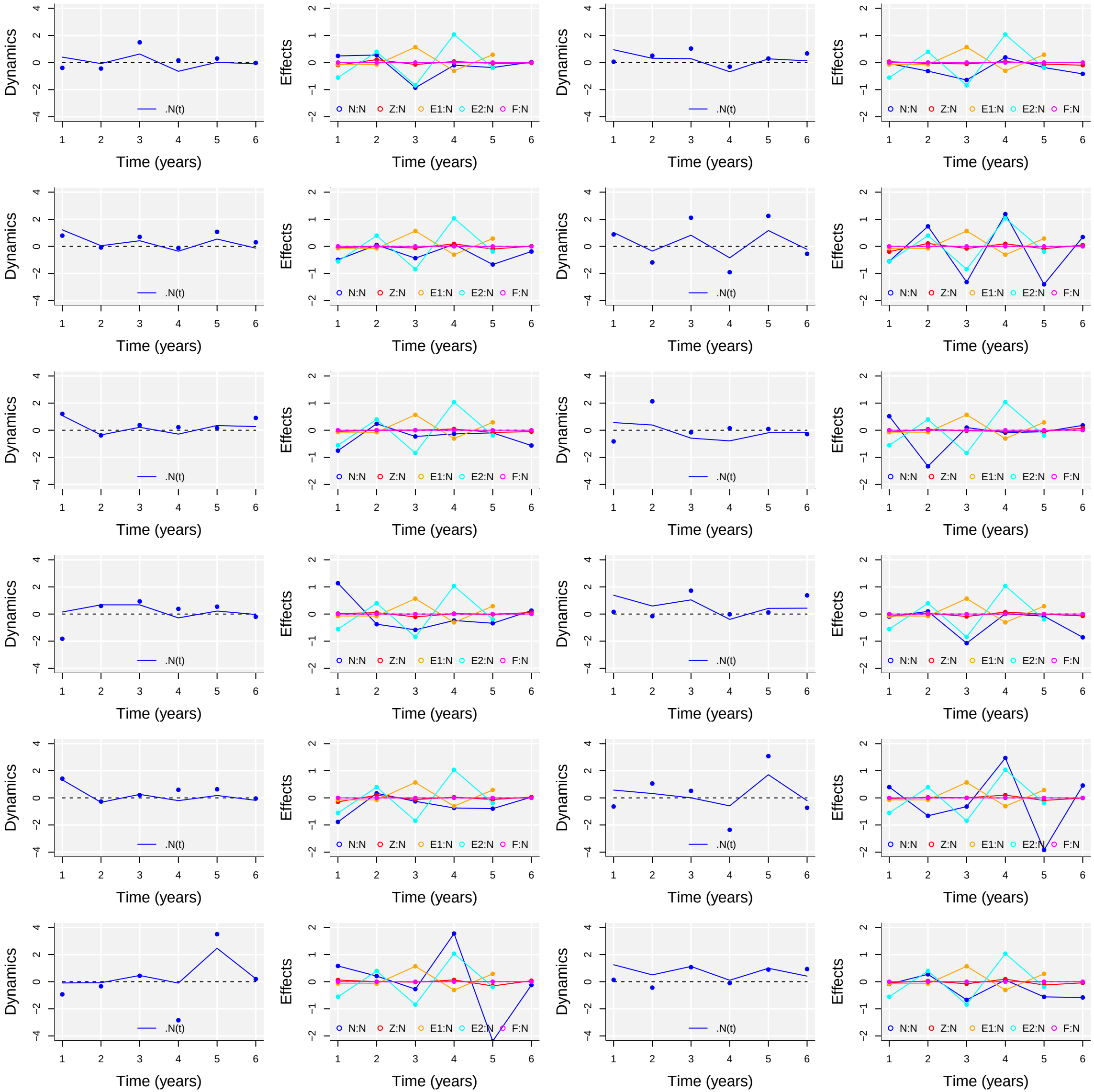

Value (S.U.)
Time (years)
N(t)
Z(t)
.N(t)
.Z(t)

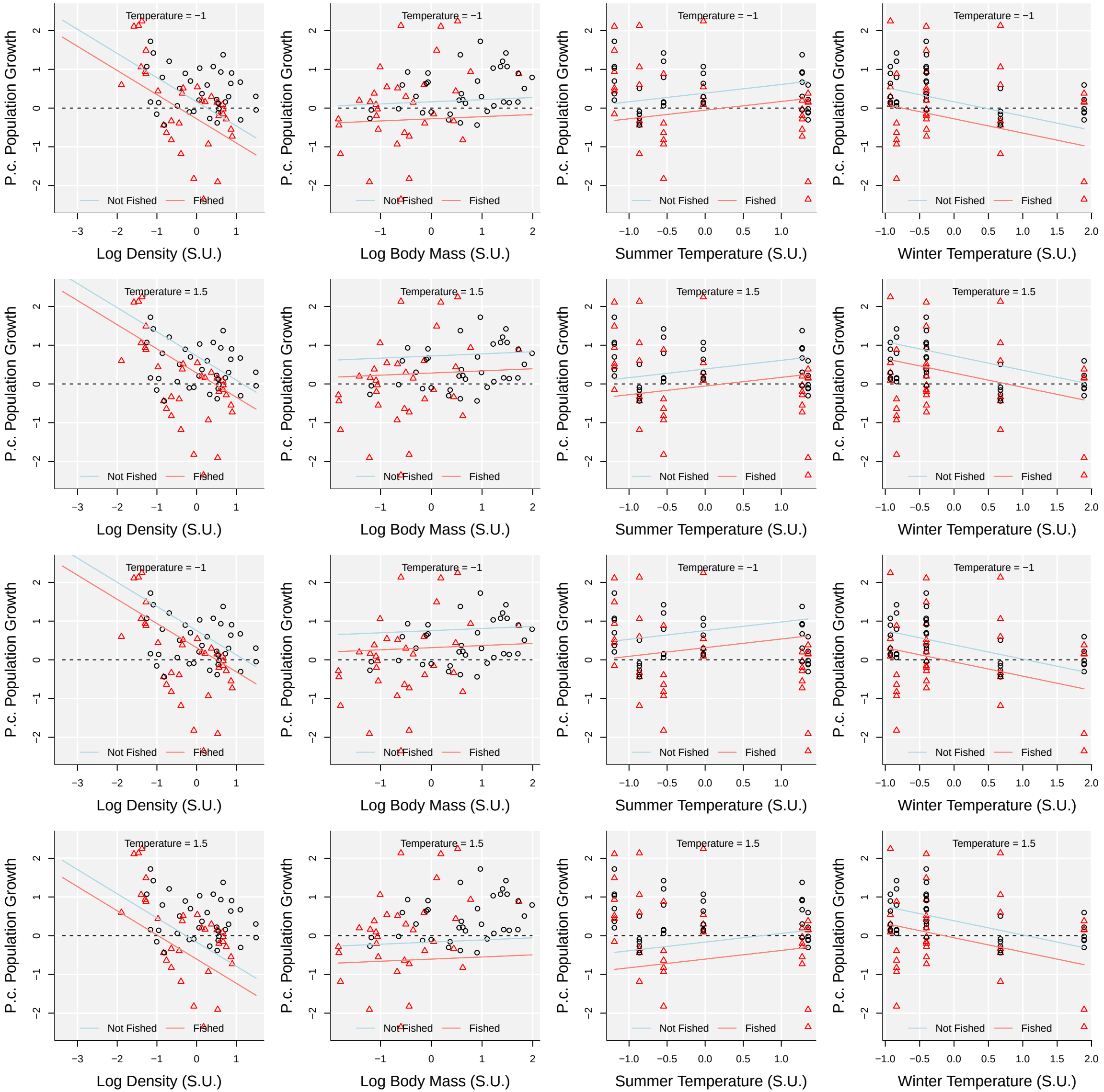
Backcast Likelihood
−2000
−2500
−3000
−3500
−4000
−4500
−5000
−5500
0.0
0.1
0.2
0.3
0.4
0.5
Regularisation Level (sd2_p)

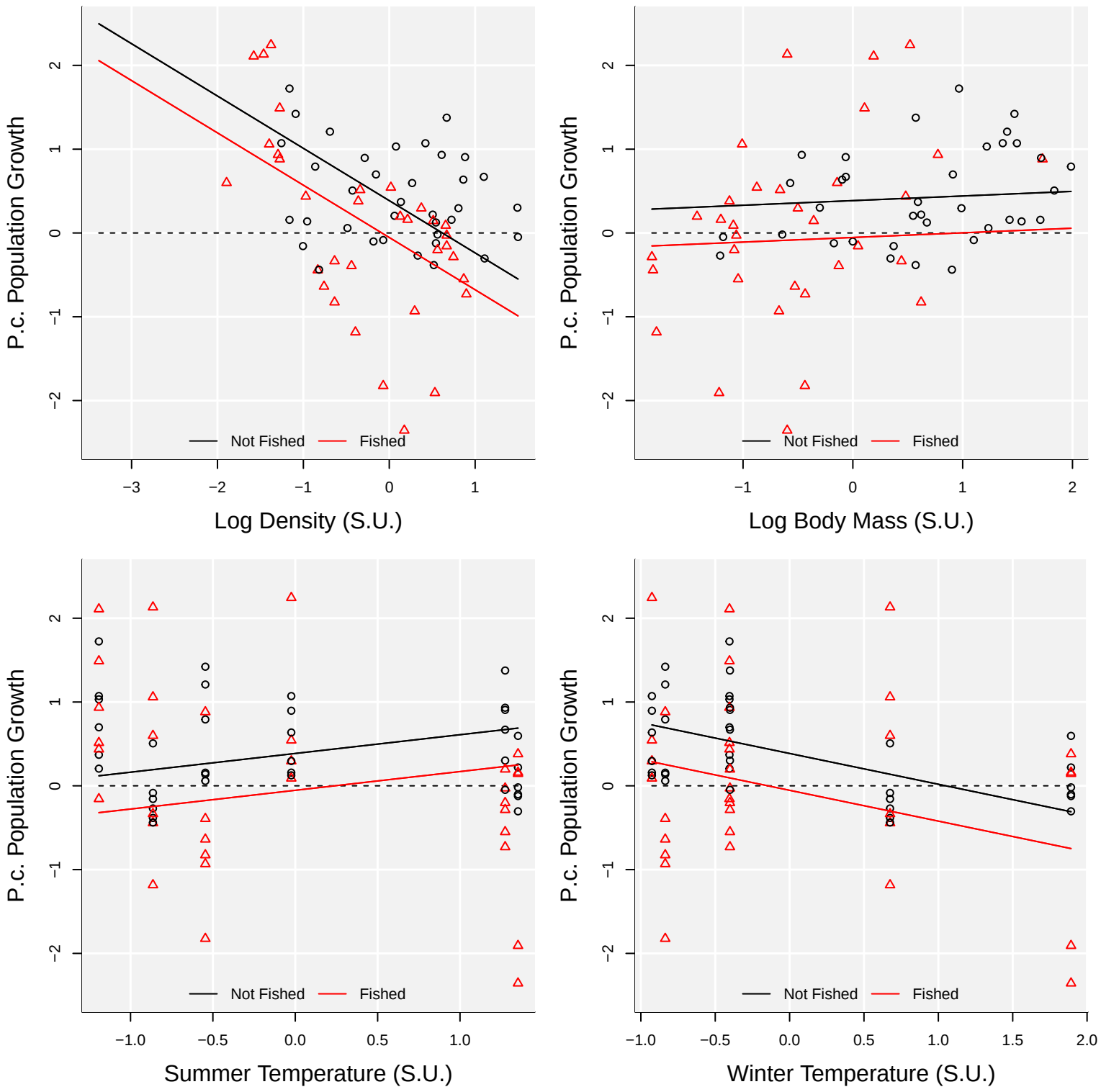

Forecast Likelihood
−2000
−4000
−6000
−8000
−10000
−12000
0.0
0.1
0.2
0.3
0.4
0.5
Regularisation Level (sd2_p)

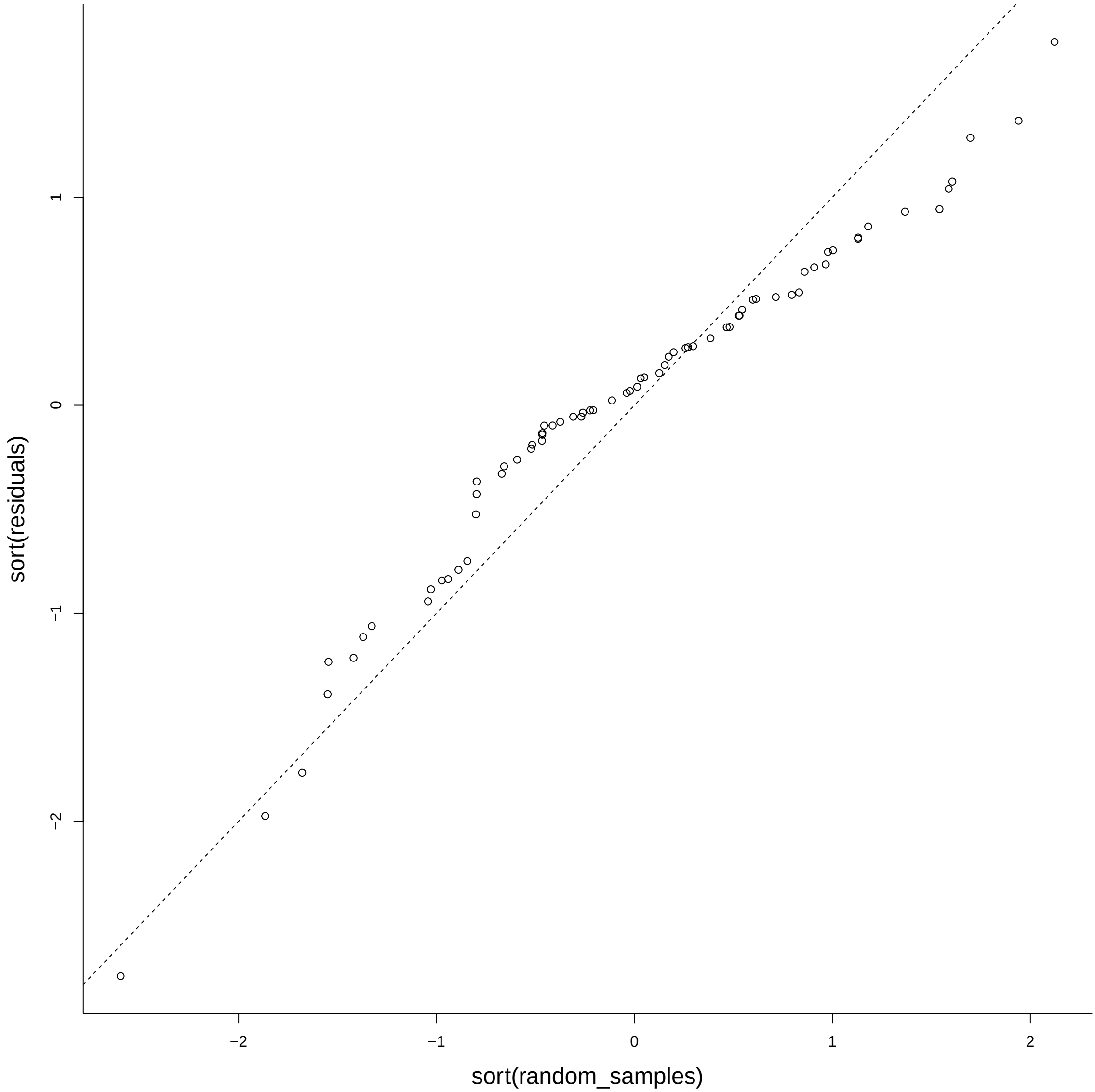

Likelihood
1.0
0.8
0.6
0.4
0.2
0.0
Backcast
Forecast
0.0
0.1
0.2
0.3
0.4
0.5
Regularisation Level (sd2_p)

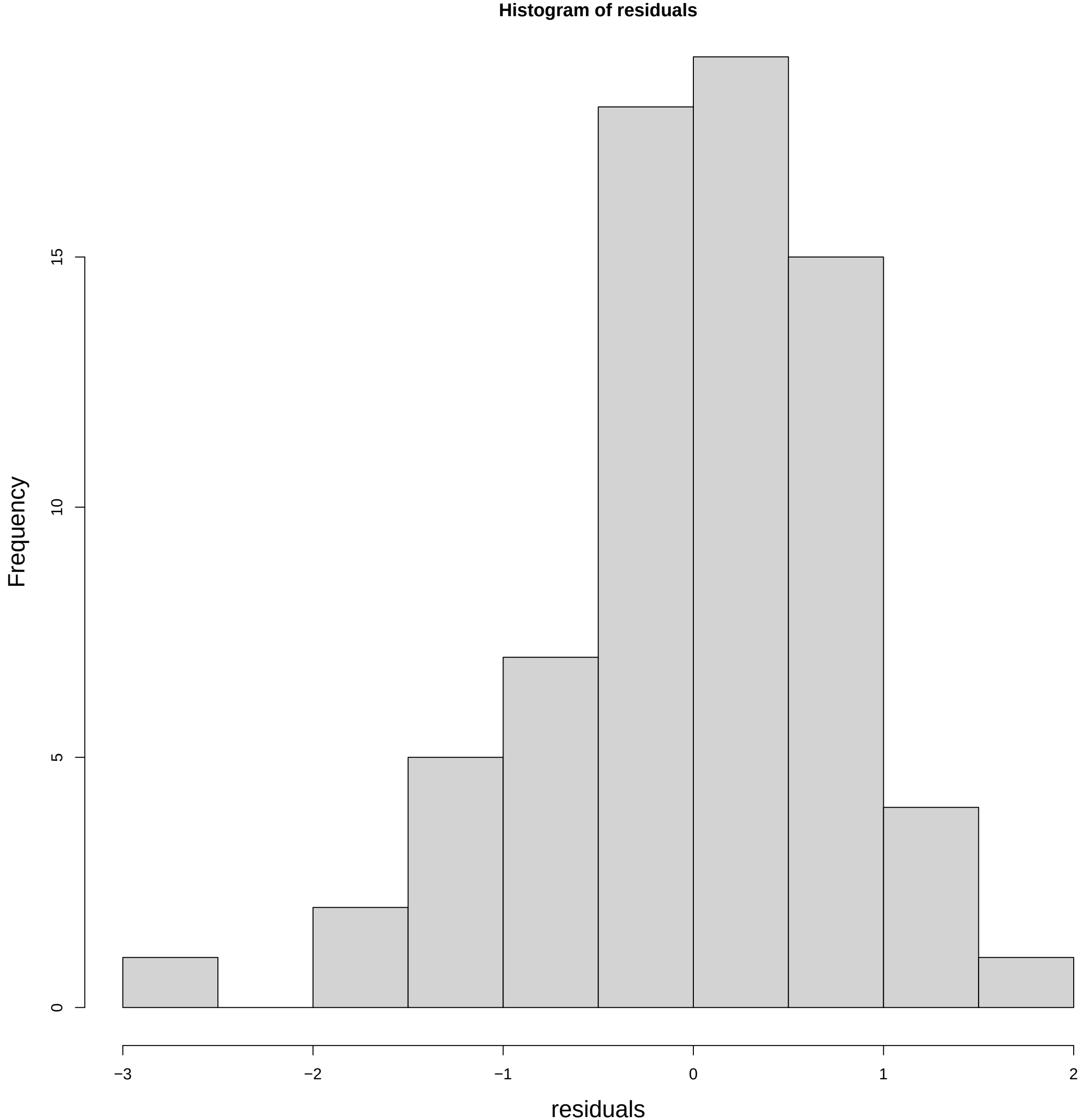

Dynamics
Effects
Time (years)
.N(t)
N:N
Z:N
E1:N
E2:N
F:N

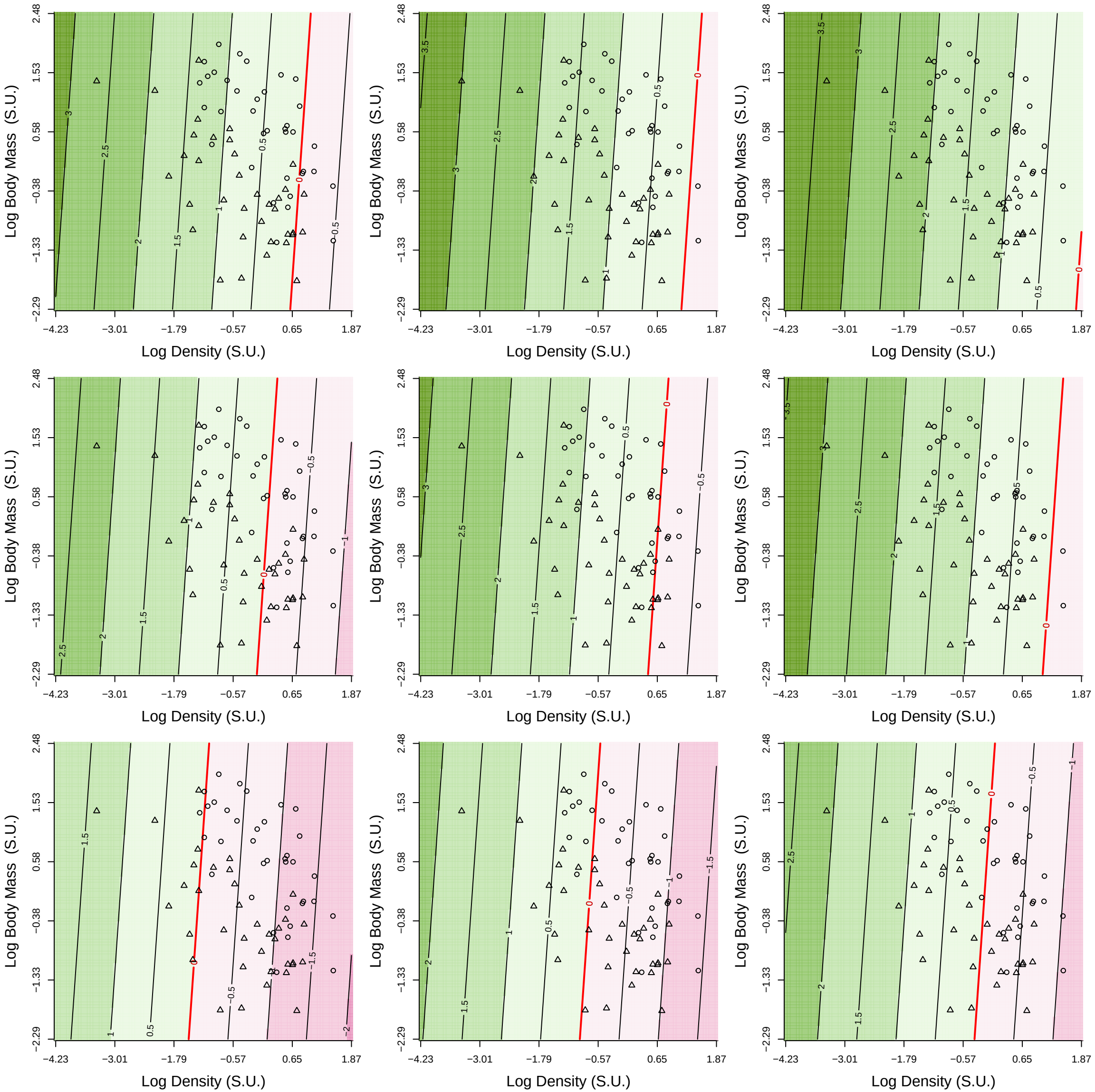

Dynamics
Effects
Time (years)
.N(t)
N:N
Z:N
E1:N
E2:N
F:N

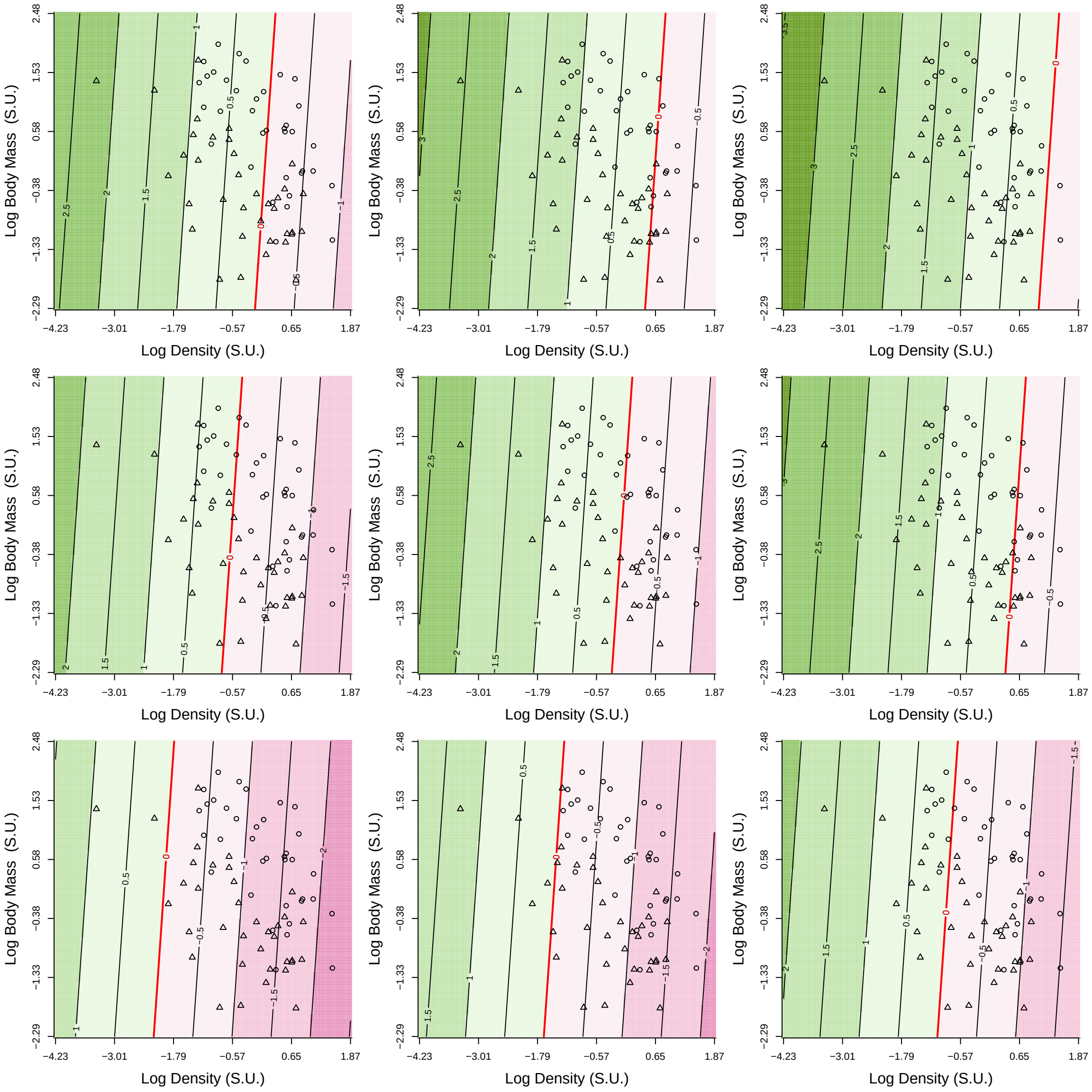

Temperature = −1
Temperature = 1.5
P.c. Population Growth
Log Density (S.U.)
Log Body Mass (S.U.)
Summer Temperature (S.U.)
Winter Temperature (S.U.)
Not Fished
Fished

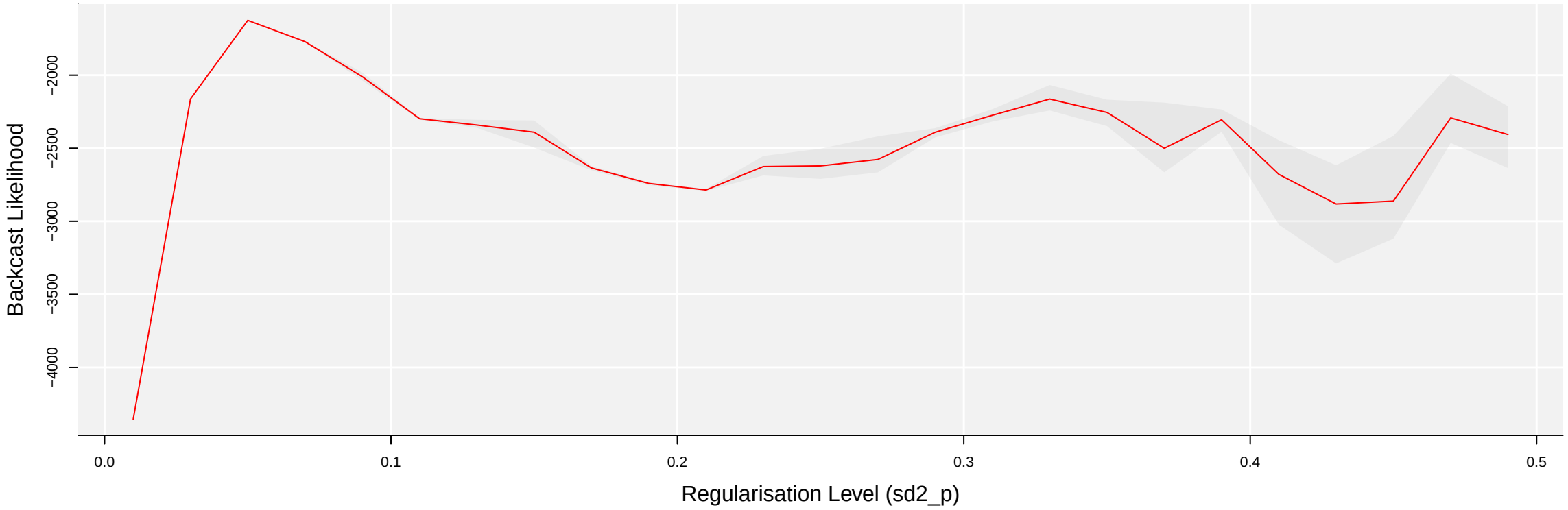

P.c. Population Growth
Log Density (S.U.)
Not Fished
Fished
P.c. Population Growth
Log Body Mass (S.U.)
Not Fished
Fished
P.c. Population Growth
Summer Temperature (S.U.)
Not Fished
Fished
P.c. Population Growth
Winter Temperature (S.U.)
Not Fished
Fished

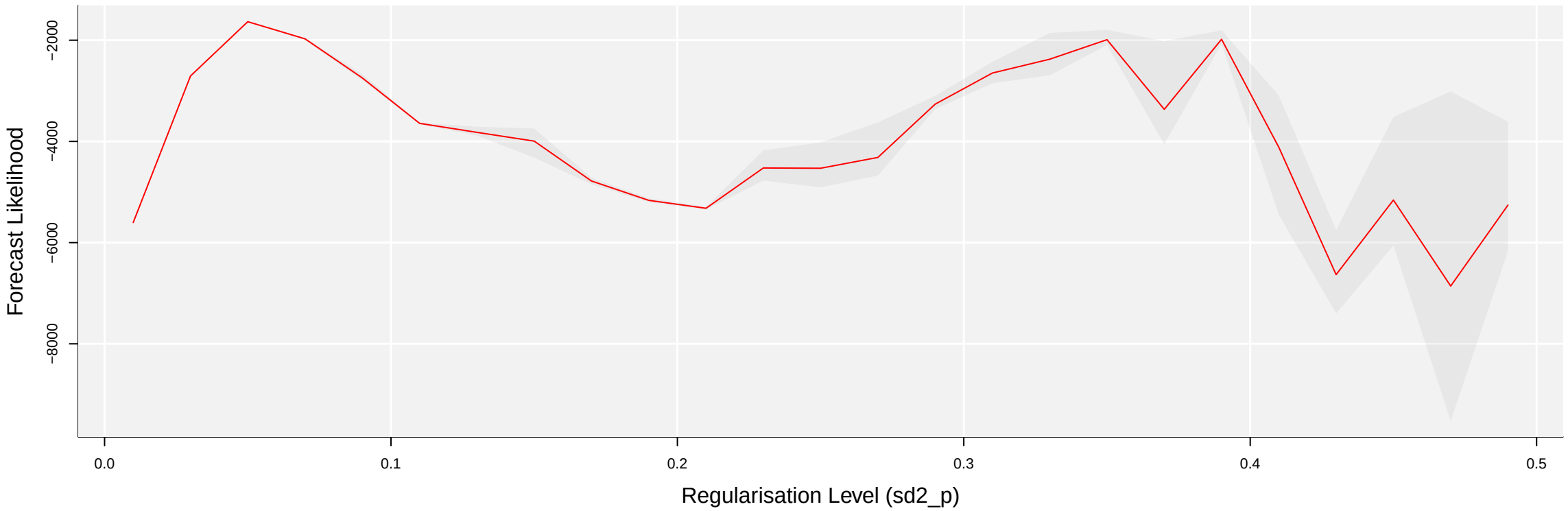

sort(residuals)
sort(random_samples)
−2
−1
0
1
2

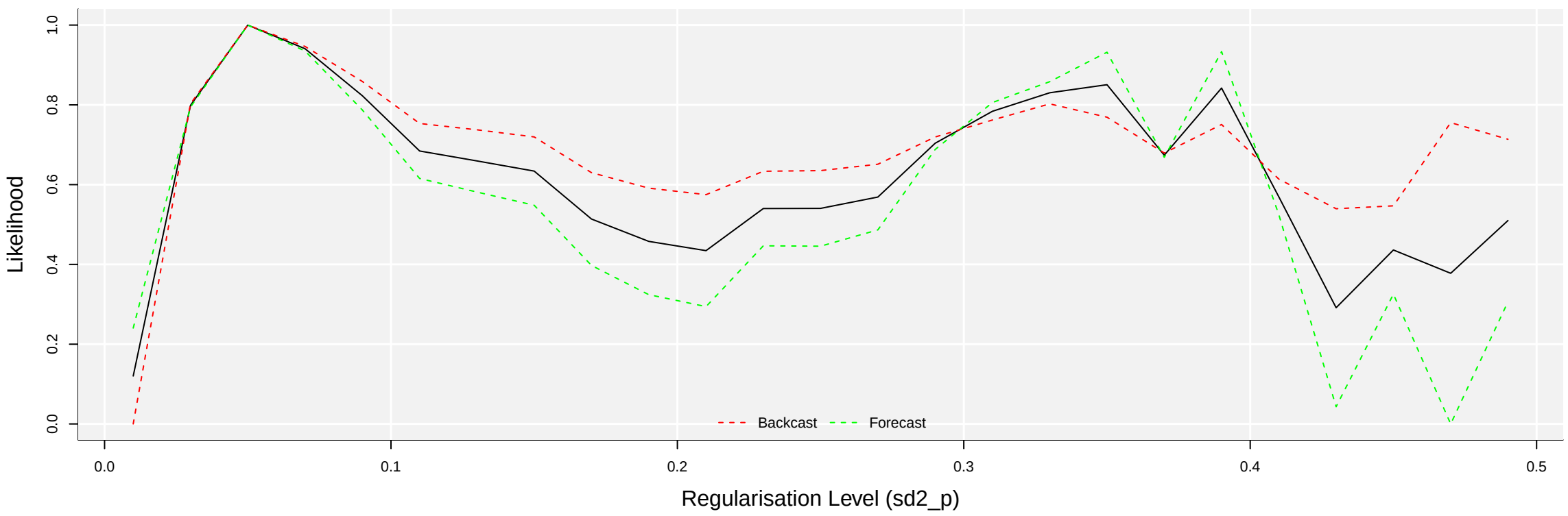

Histogram of residuals
Frequency
0
5
10
15
−3
−2
−1
0
1
2
residuals

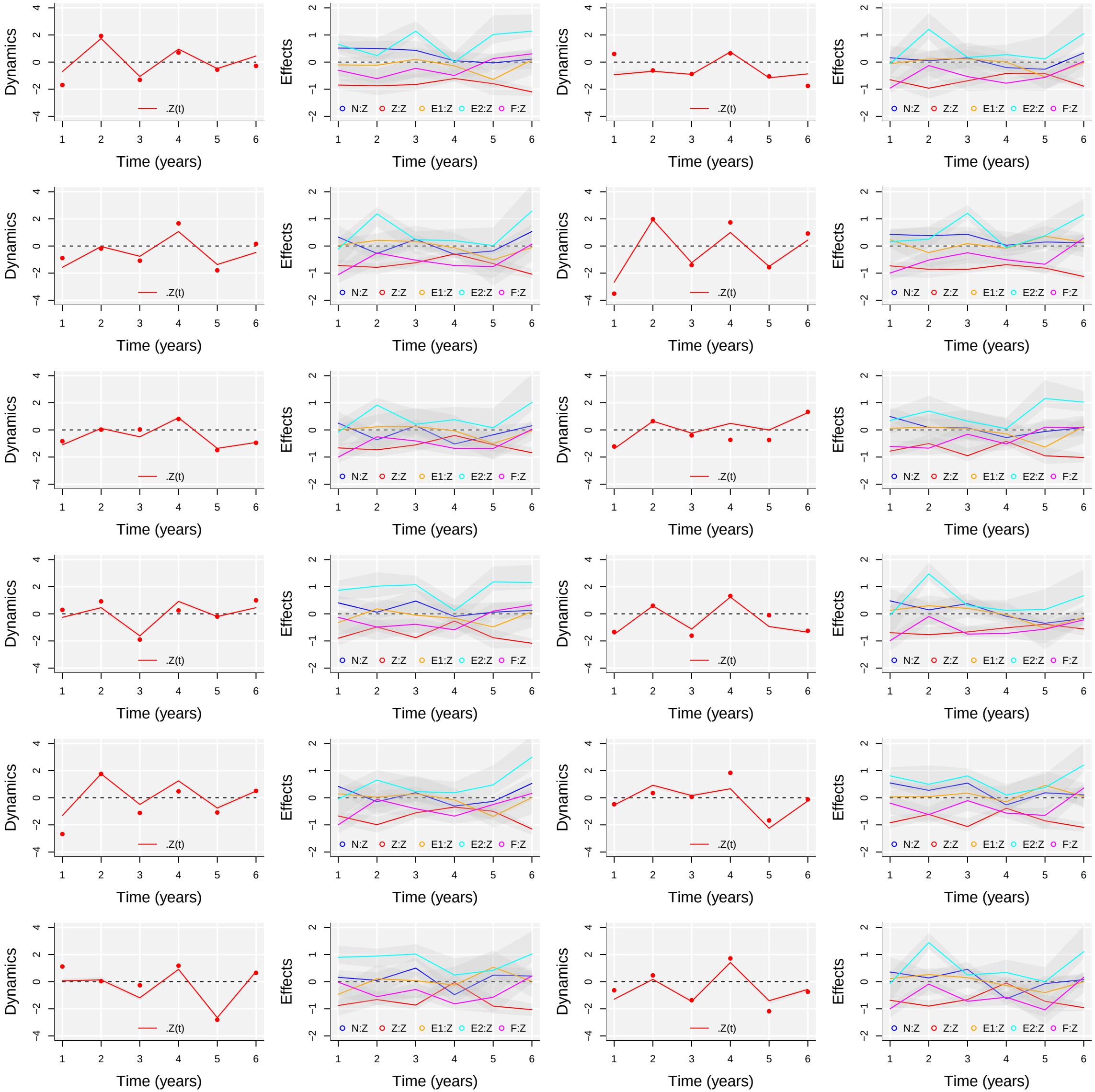

Log Body Mass (S.U.)
Log Density (S.U.)
2.48
1.53
0.58
−0.38
−1.33
−2.29
−4.23
−3.01
−1.79
−0.57
0.65
1.87

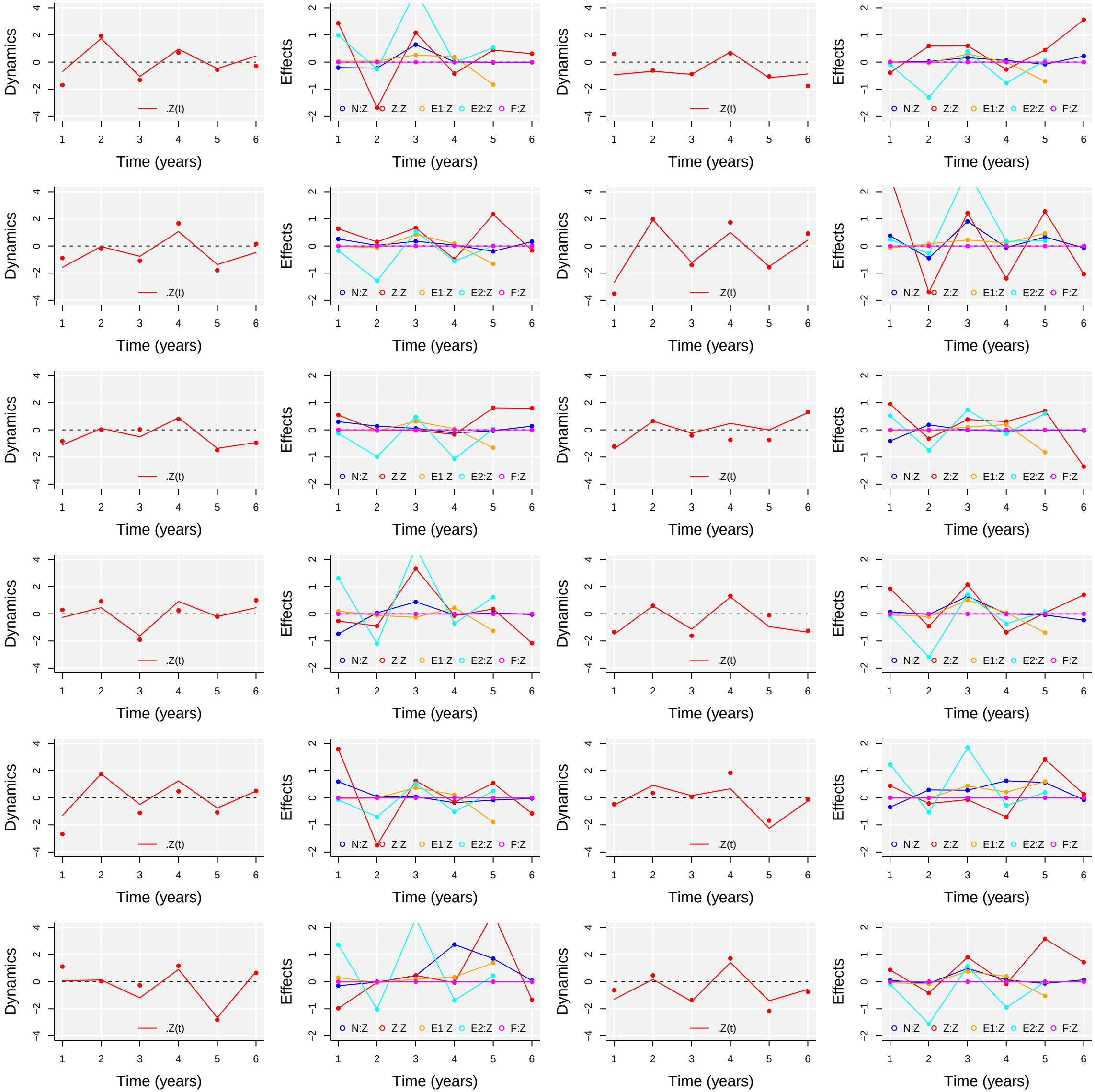

Log Body Mass (S.U.)
Log Density (S.U.)
2.48
1.53
0.58
−0.38
−1.33
−2.29
−4.23
−3.01
−1.79
−0.57
0.65
1.87

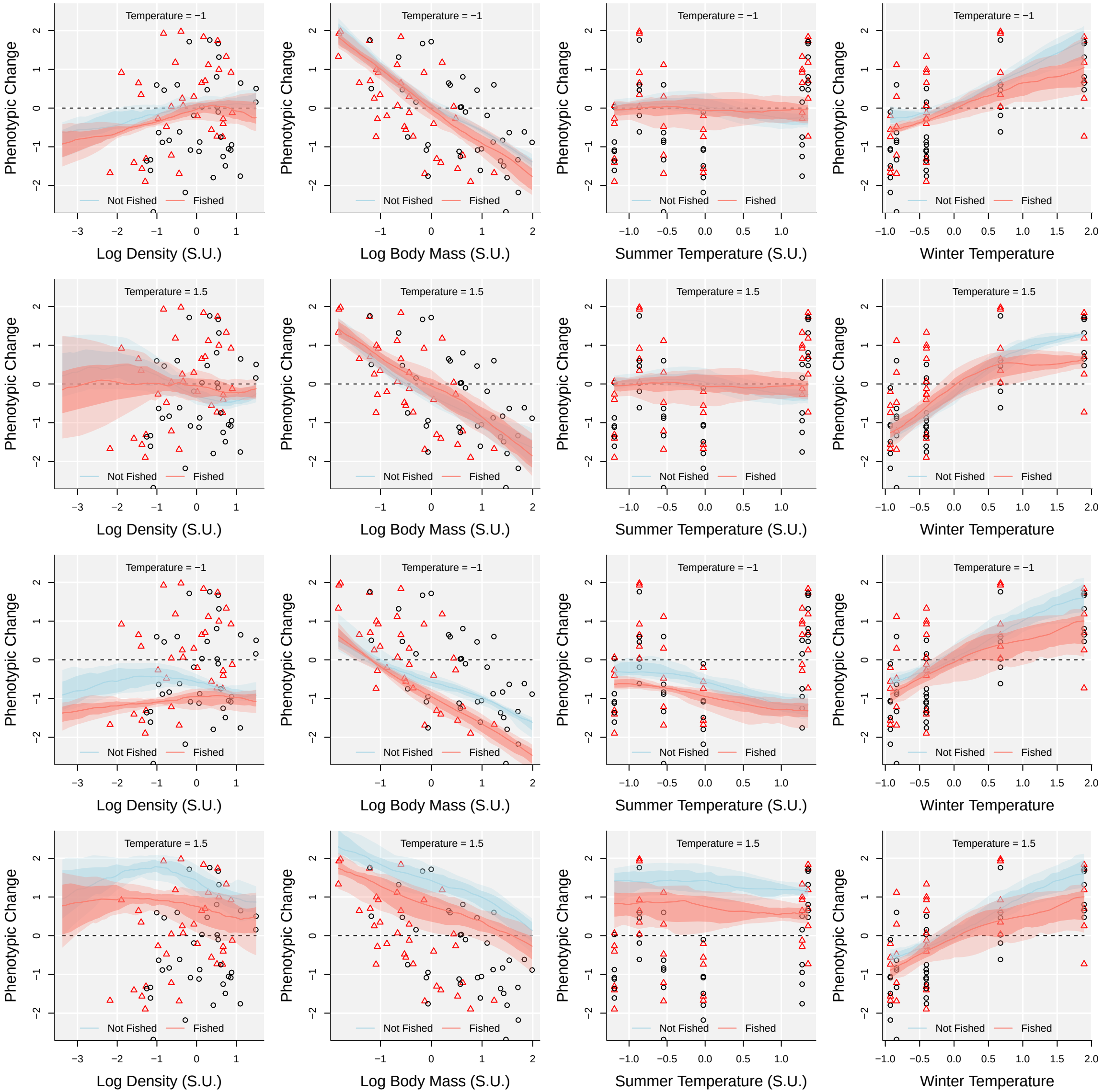

Backcast Likelihood
−2000
−2500
−3000
−3500
−4000
0.0
0.1
0.2
0.3
0.4
0.5
Regularisation Level (sd2_p)

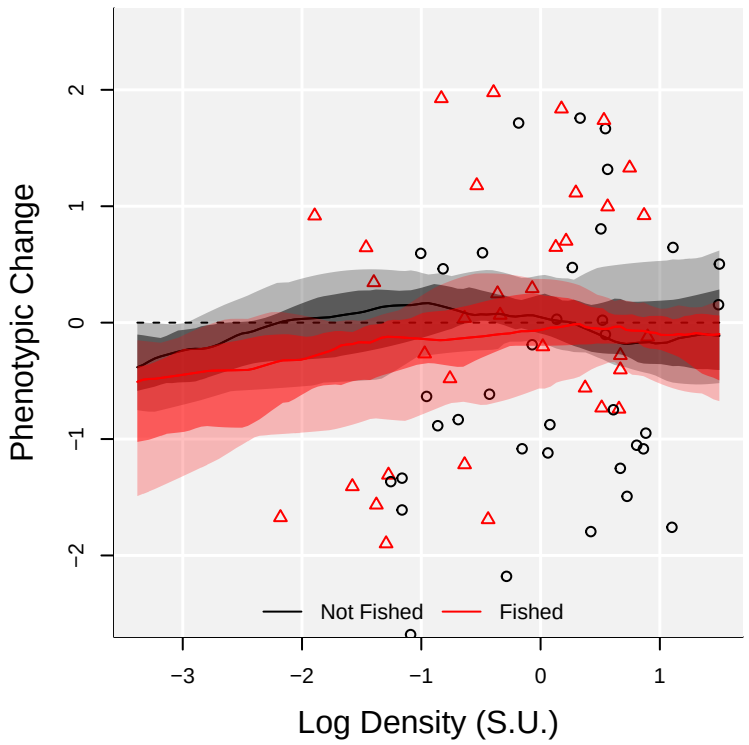

Forecast Likelihood
−2000
−4000
−6000
−8000
0.0
0.1
0.2
0.3
0.4
0.5
Regularisation Level (sd2_p)

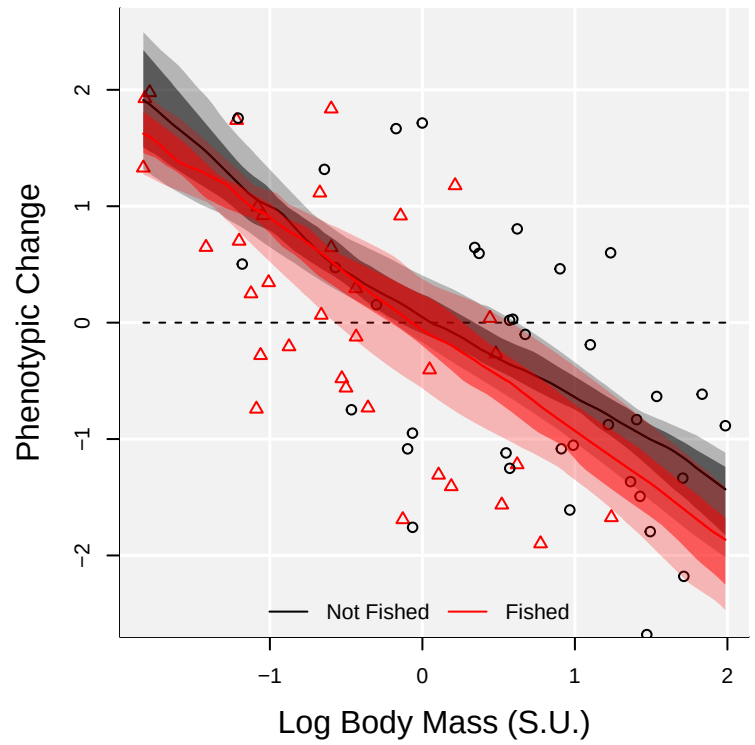

Likelihood
1.0
0.8
0.6
0.4
0.2
0.0
Backcast
Forecast
0.0
0.1
0.2
0.3
0.4
0.5
Regularisation Level (sd2_p)

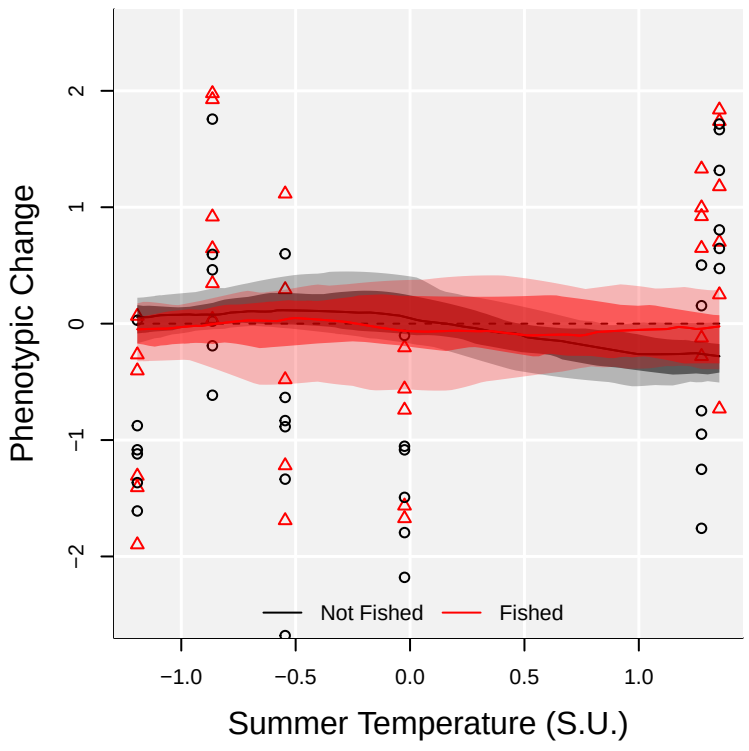

Dynamics
Effects
Time (years)
.Z(t)
N:Z
Z:Z
E1:Z
E2:Z
F:Z

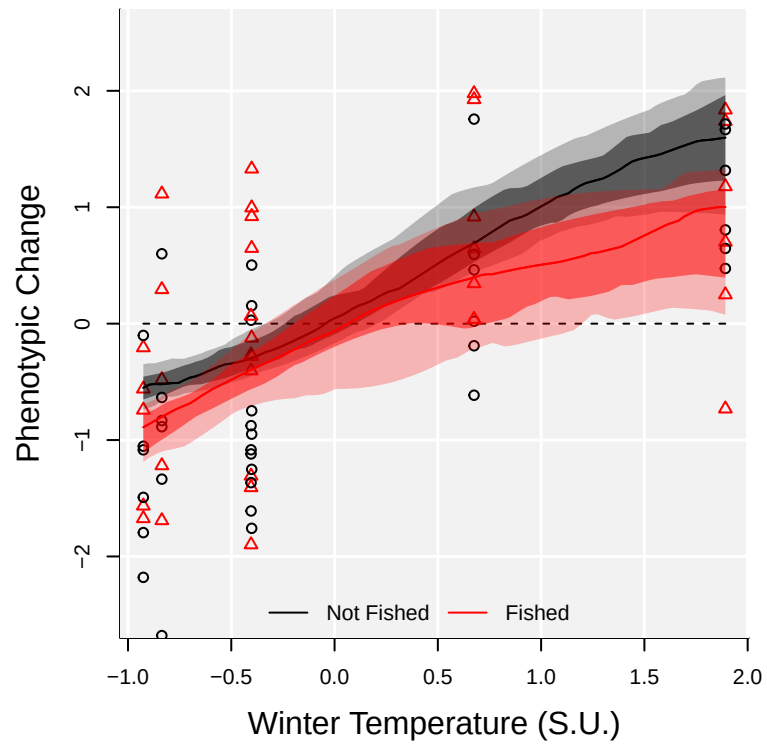

Dynamics
Effects
Time (years)
.Z(t)
N:Z
Z:Z
E1:Z
E2:Z
F:Z

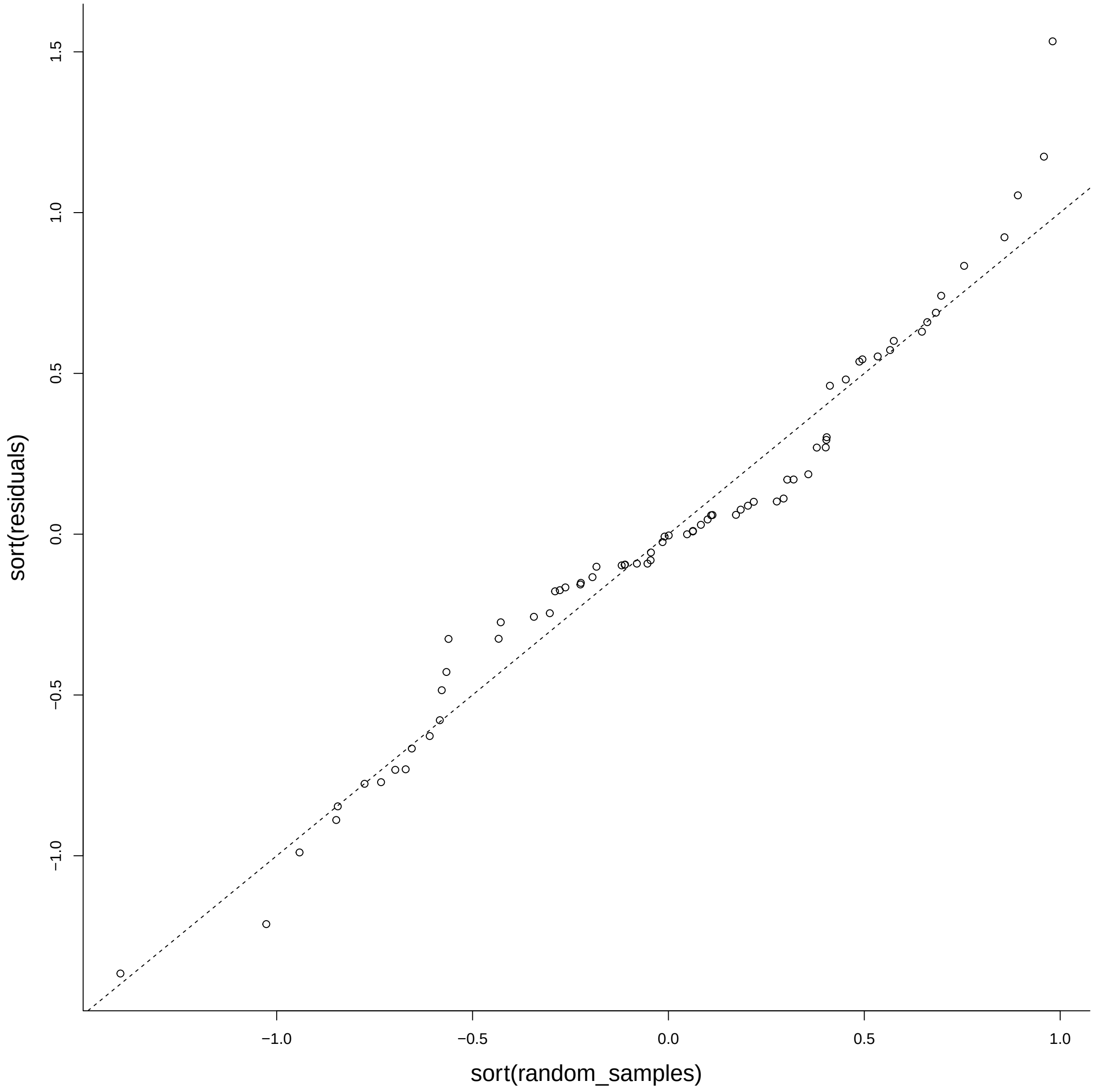

Temperature = −1
Temperature = 1.5
Phenotypic Change
Not Fished
Fished
Log Density (S.U.)
Log Body Mass (S.U.)
Summer Temperature (S.U.)
Winter Temperature

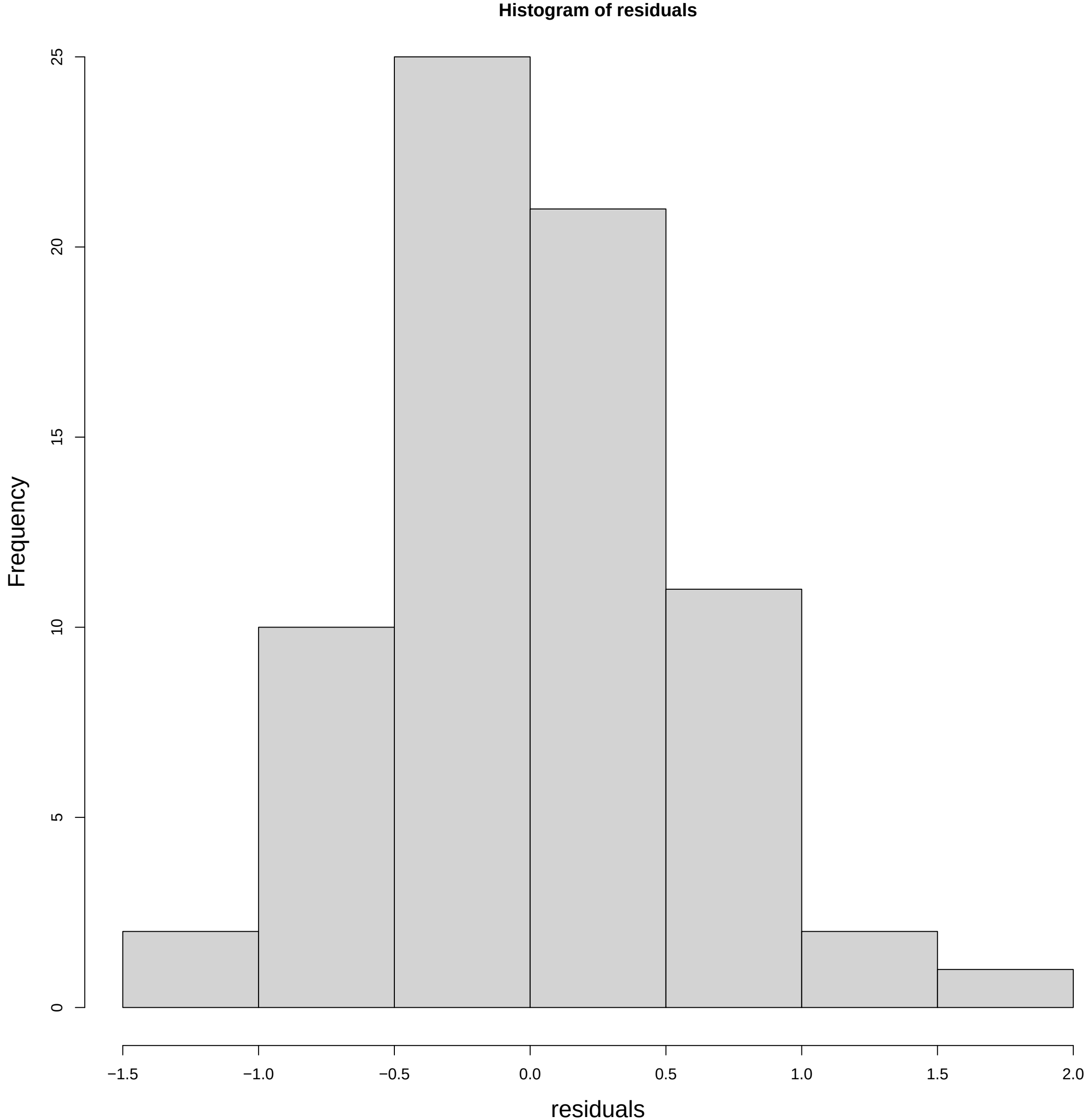

Phenotypic Change
Log Density (S.U.)
Not Fished
Fished

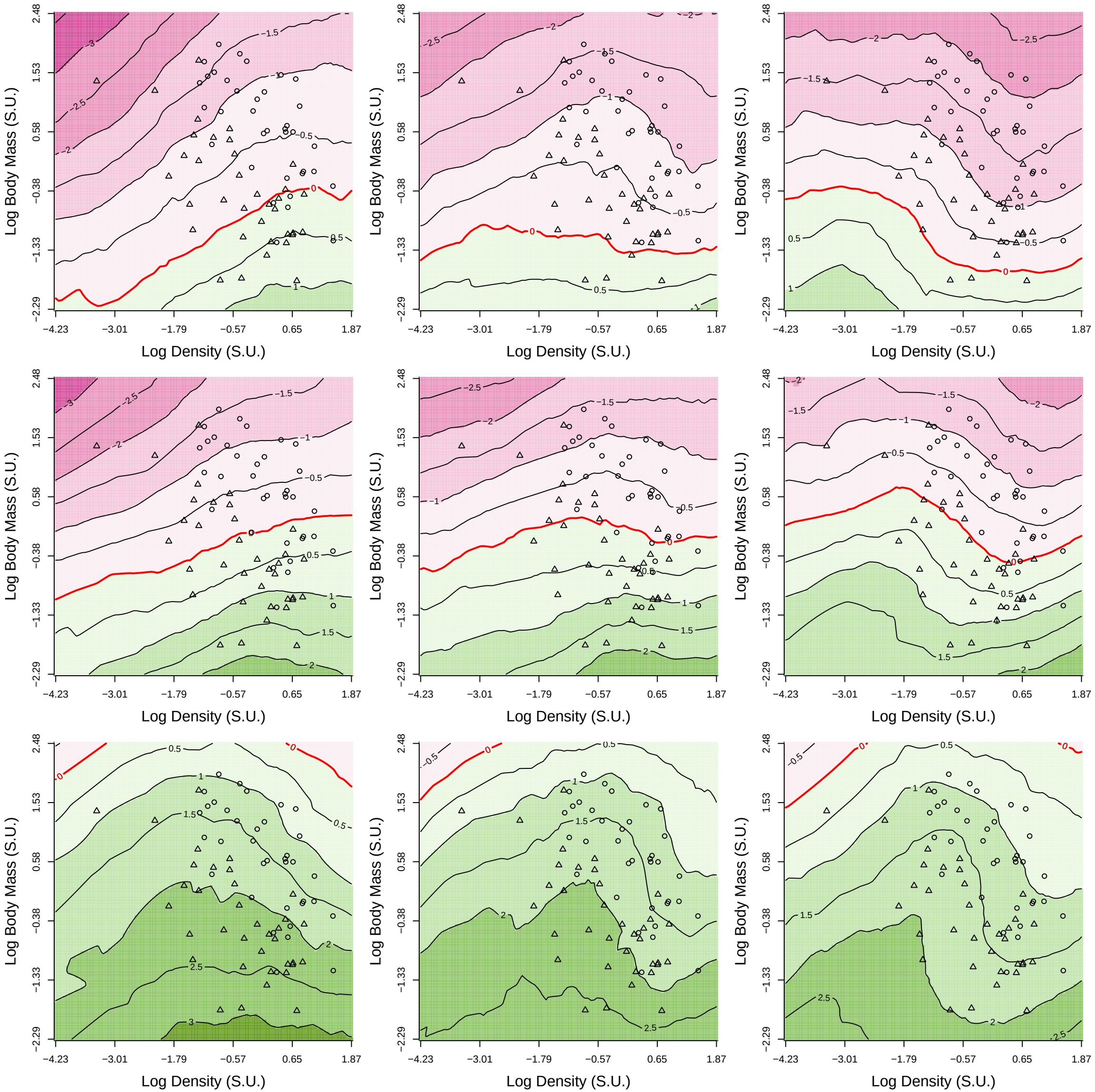

Phenotypic Change
Log Body Mass (S.U.)
Not Fished
Fished

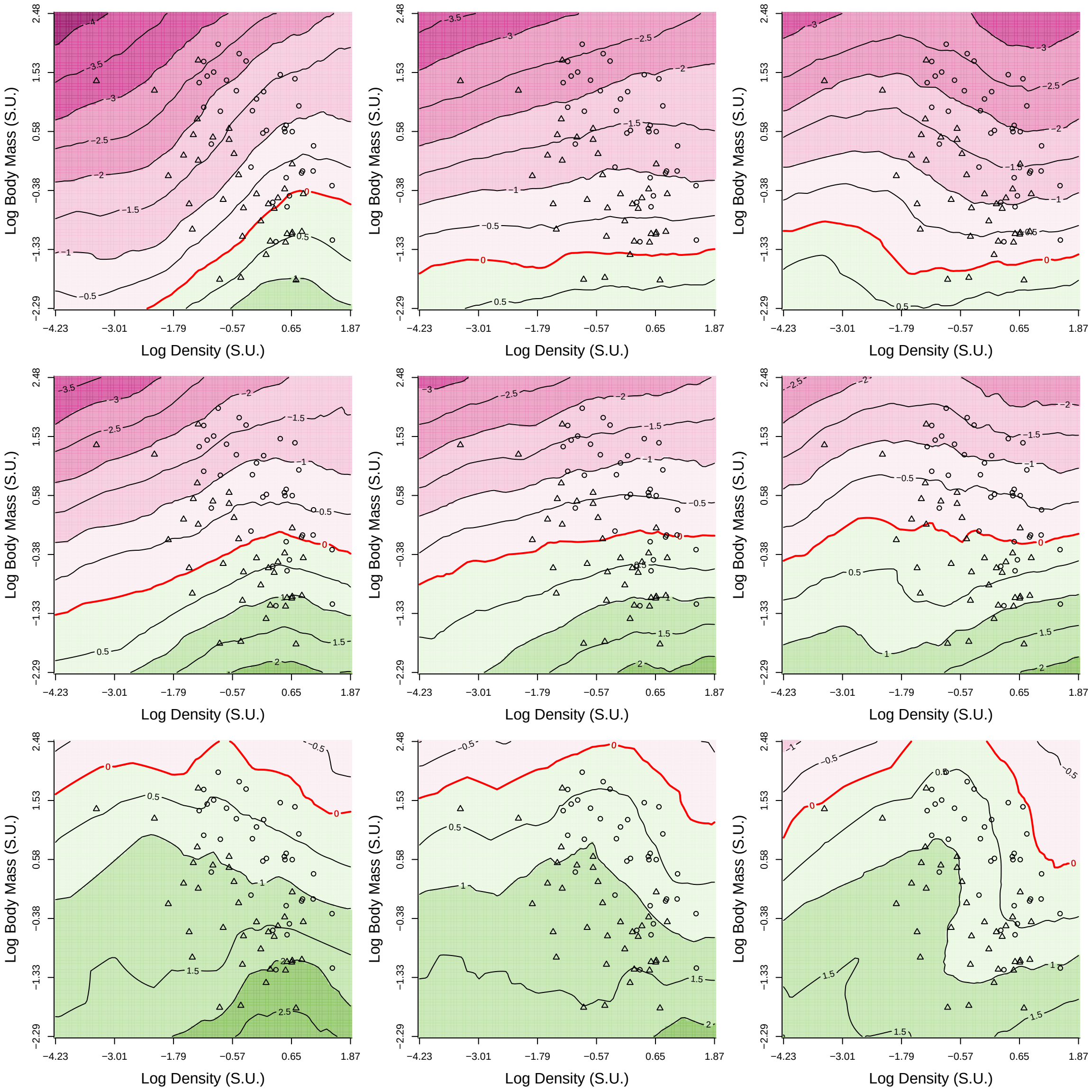

Phenotypic Change
Summer Temperature (S.U.)
Not Fished
Fished

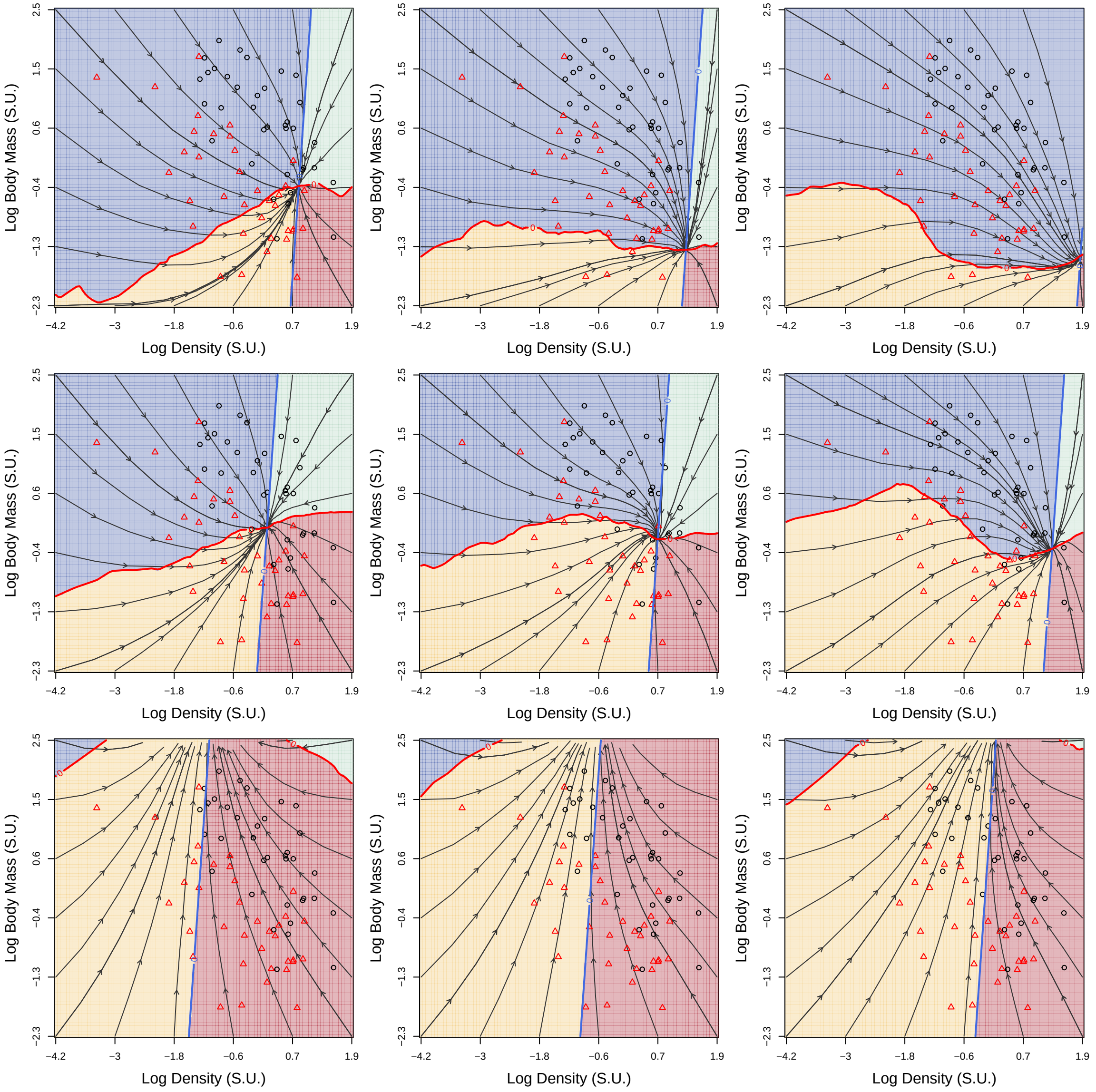

Phenotypic Change
Winter Temperature (S.U.)
Not Fished
Fished

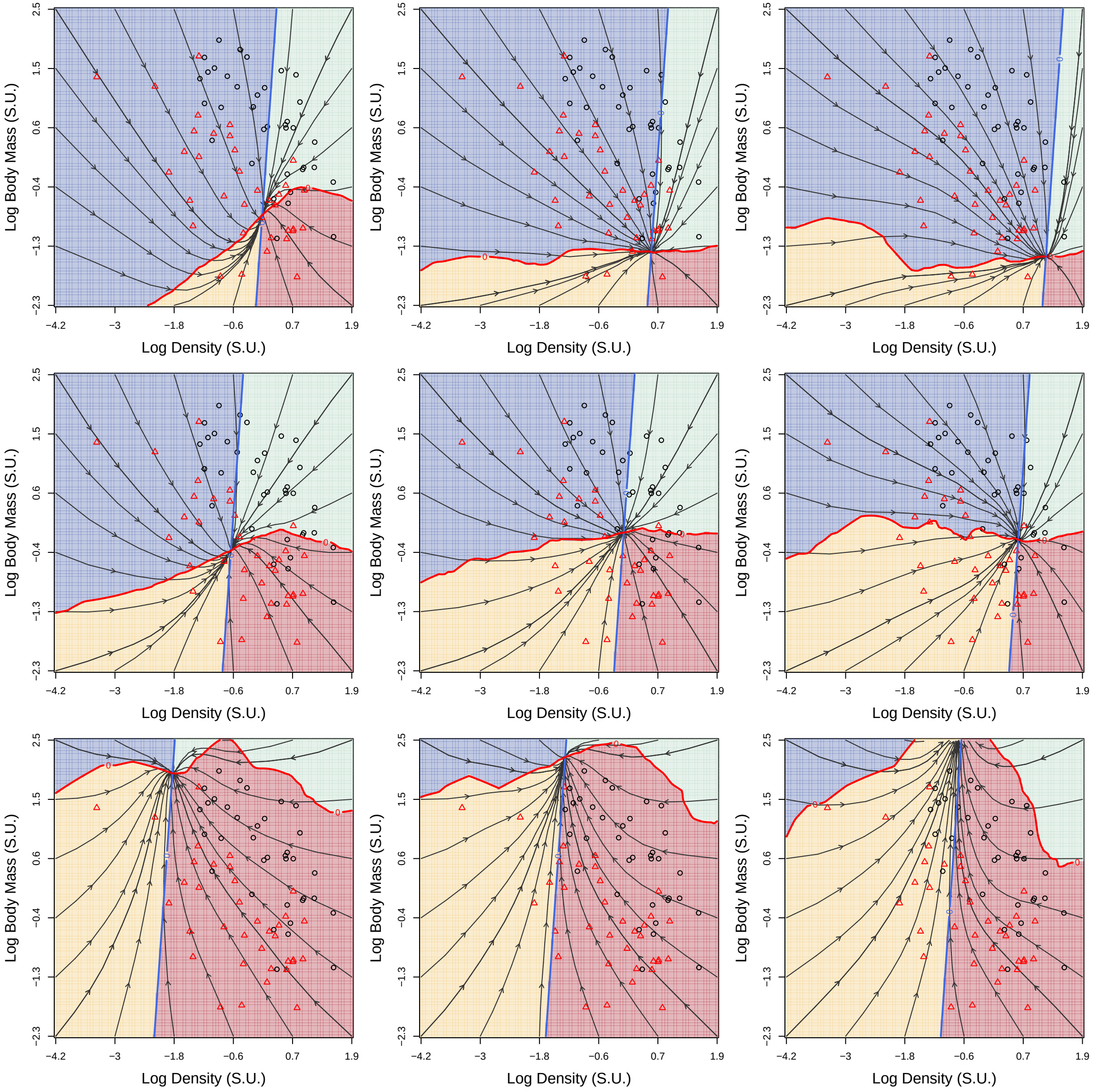

sort(residuals)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
sort(random_samples)

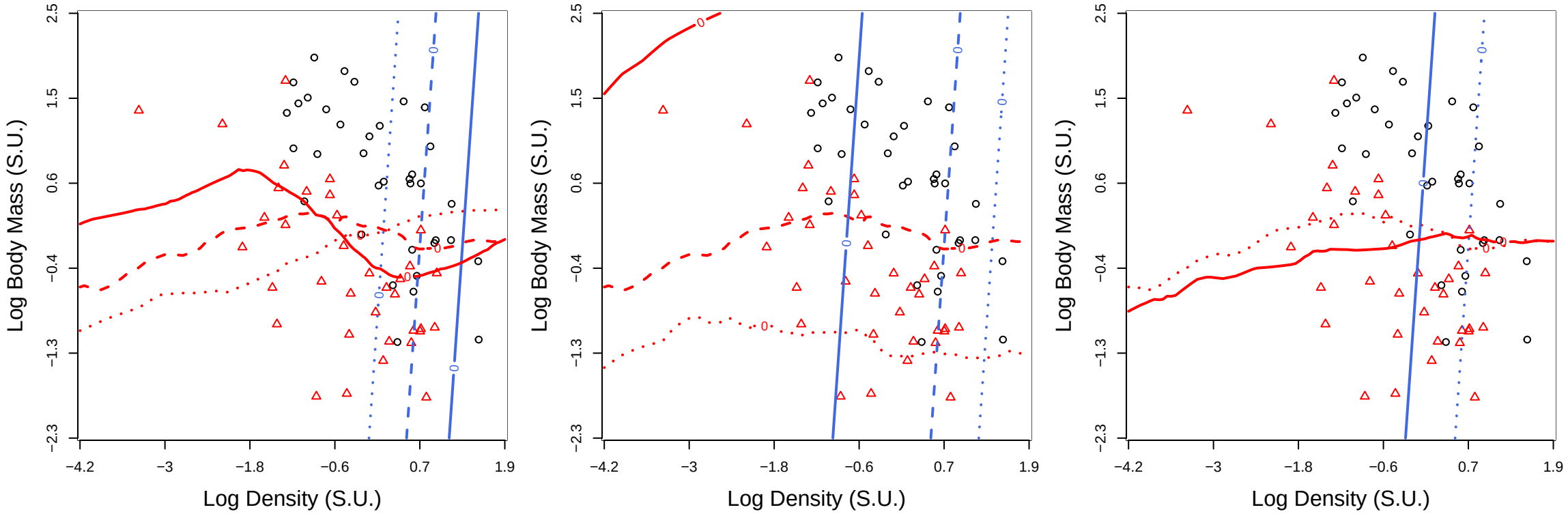

Histogram of residuals
Frequency
0
5
10
15
20
25
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
residuals

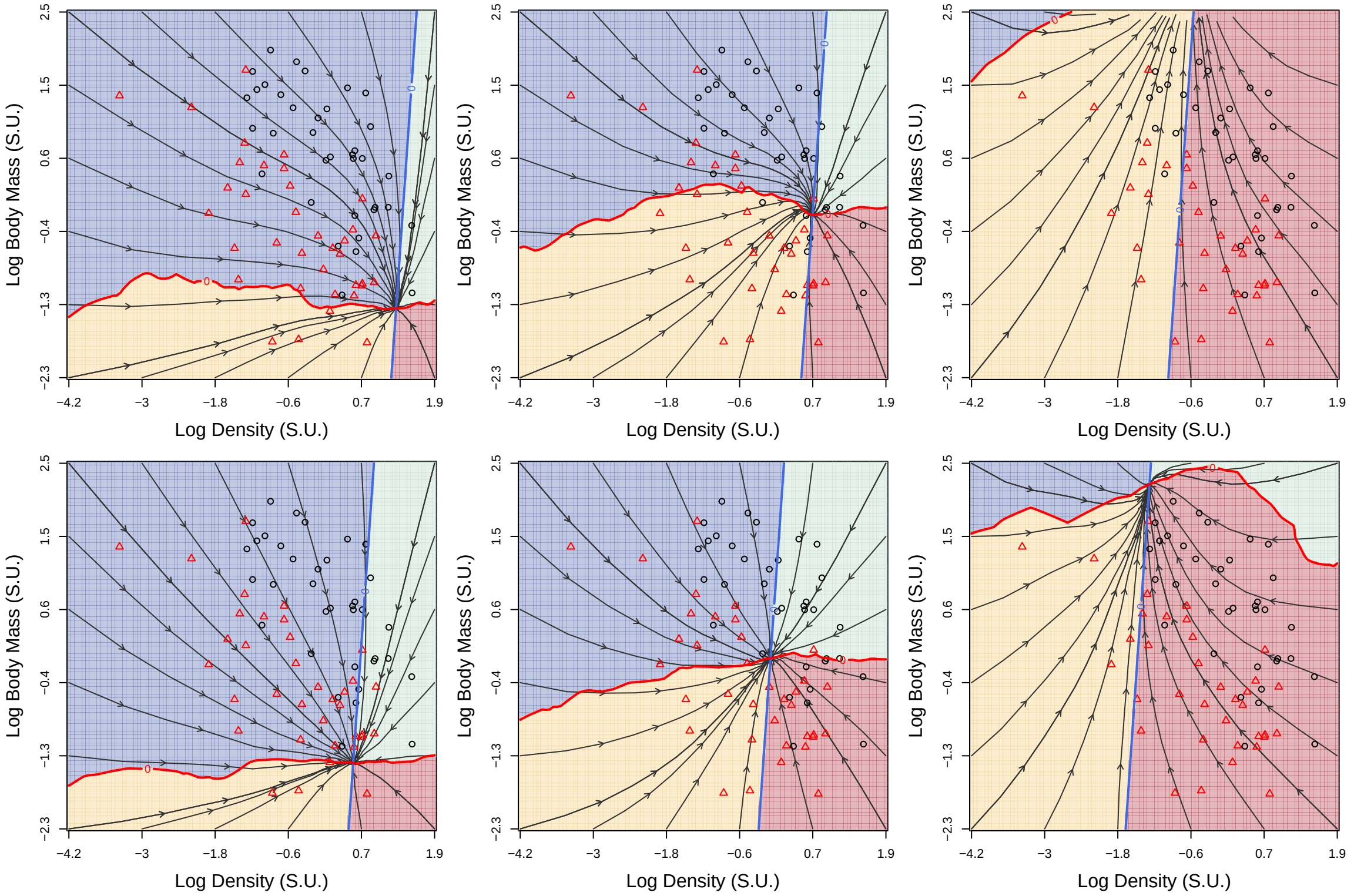

Log Body Mass (S.U.)
Log Density (S.U.)
2.48
1.53
0.58
−0.38
−1.33
−2.29
−4.23
−3.01
−1.79
−0.57
0.65
1.87


Log Body Mass (S.U.)
Log Density (S.U.)
2.48
1.53
0.58
−0.38
−1.33
−2.29
−4.23
−3.01
−1.79
−0.57
0.65
1.87


Log Body Mass (S.U.)
Log Density (S.U.)
2.5
1.5
0.6
−0.4
−1.3
−2.3
−4.2
−3
−1.8
−0.6
0.7
1.9
0


Log Body Mass (S.U.)
Log Density (S.U.)
2.5
1.5
0.6
−0.4
−1.3
−2.3
−4.2
−3
−1.8
−0.6
0.7
1.9
0


Log Body Mass (S.U.)
Log Density (S.U.)
2.5
1.5
0.6
−0.4
−1.3
−2.3
−4.2
−3
−1.8
−0.6
0.7
1.9


Log Body Mass (S.U.)
Log Density (S.U.)
2.5
1.5
0.6
−0.4
−1.3
−2.3
−4.2
−3
−1.8
−0.6
0.7
1.9
0